Bayesian Inference and Hypothesis Testing

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Outline

Introduction

Conditional Probability and the Theorem of Bayes

Simple Bayesian hypothesis testing

Bayesian Inference

Exercises

Introduction

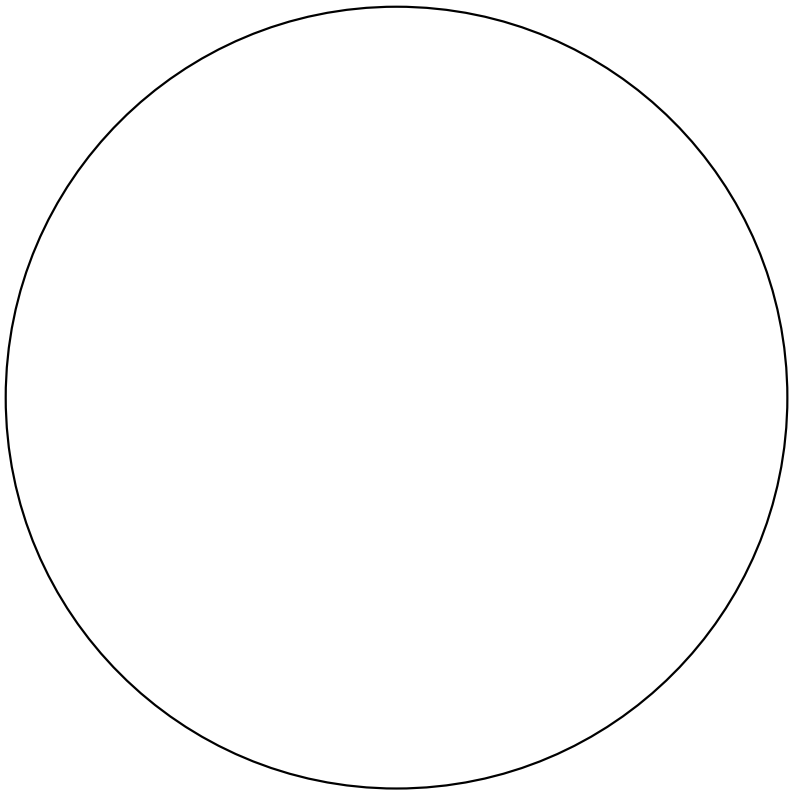
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Probability refresher

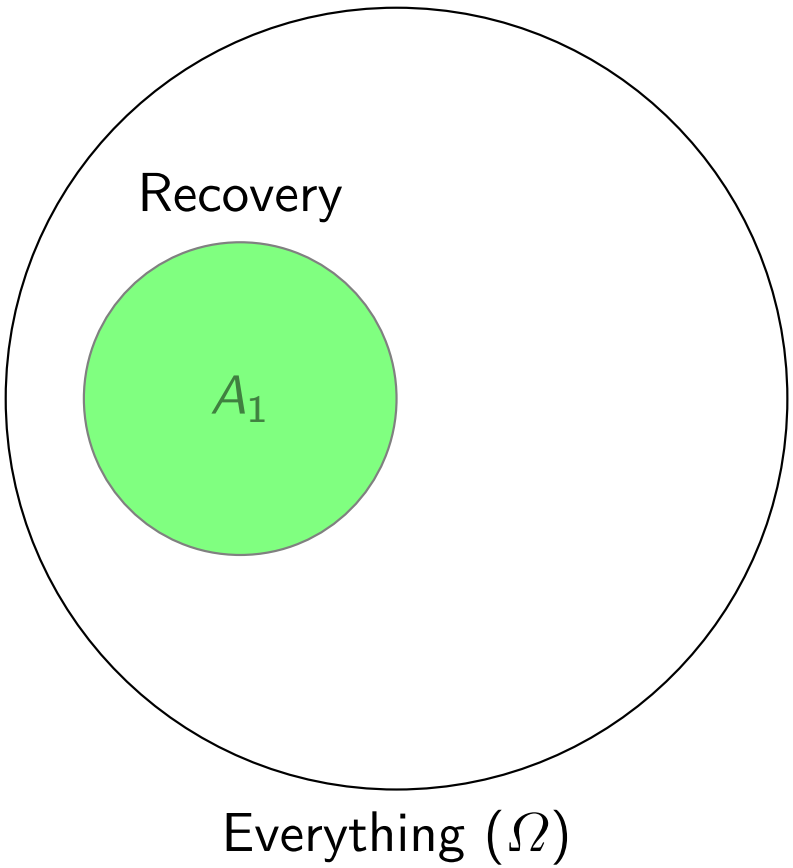


Everything (Ω)

Notation

- $\omega \in \Omega$ random outcome

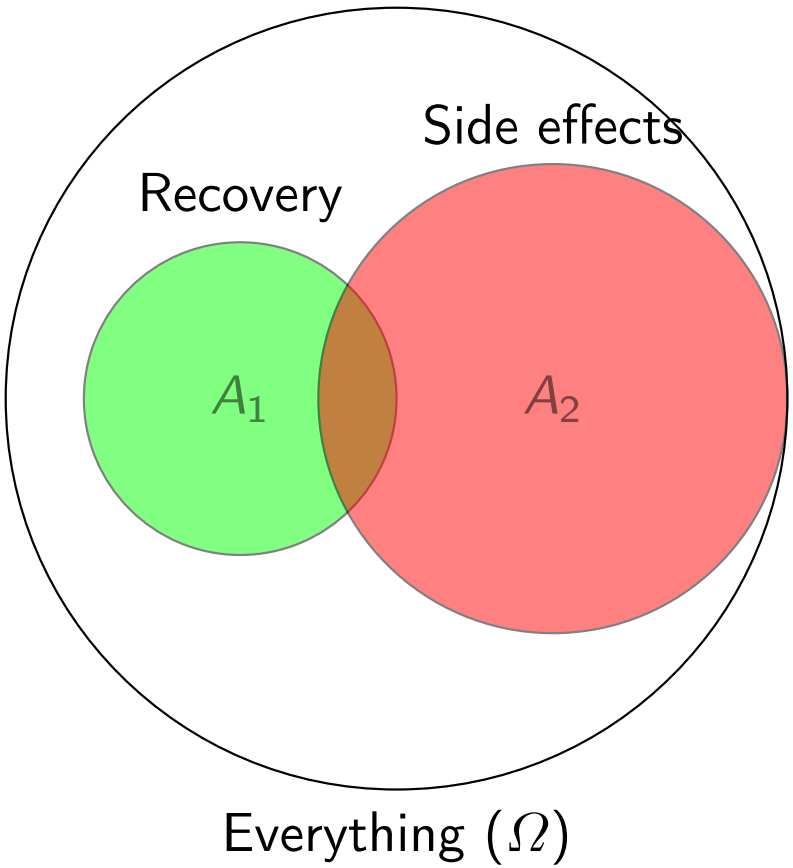
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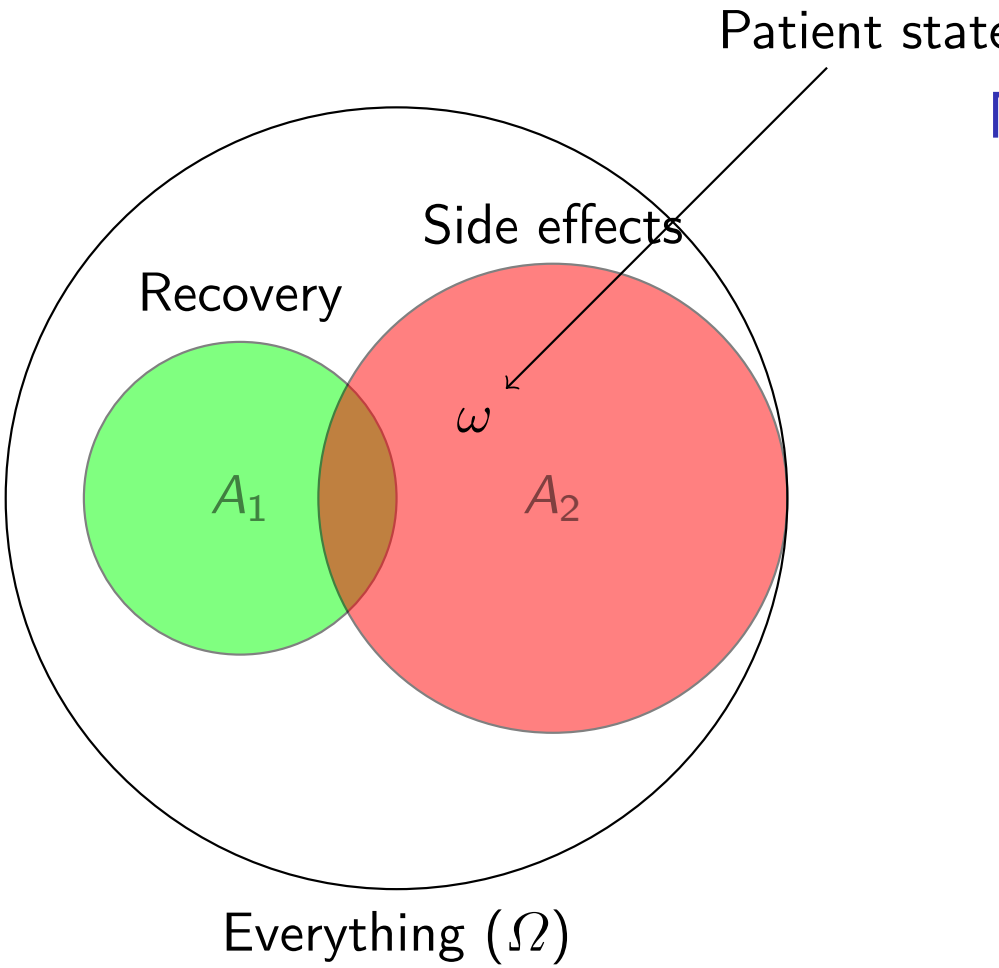
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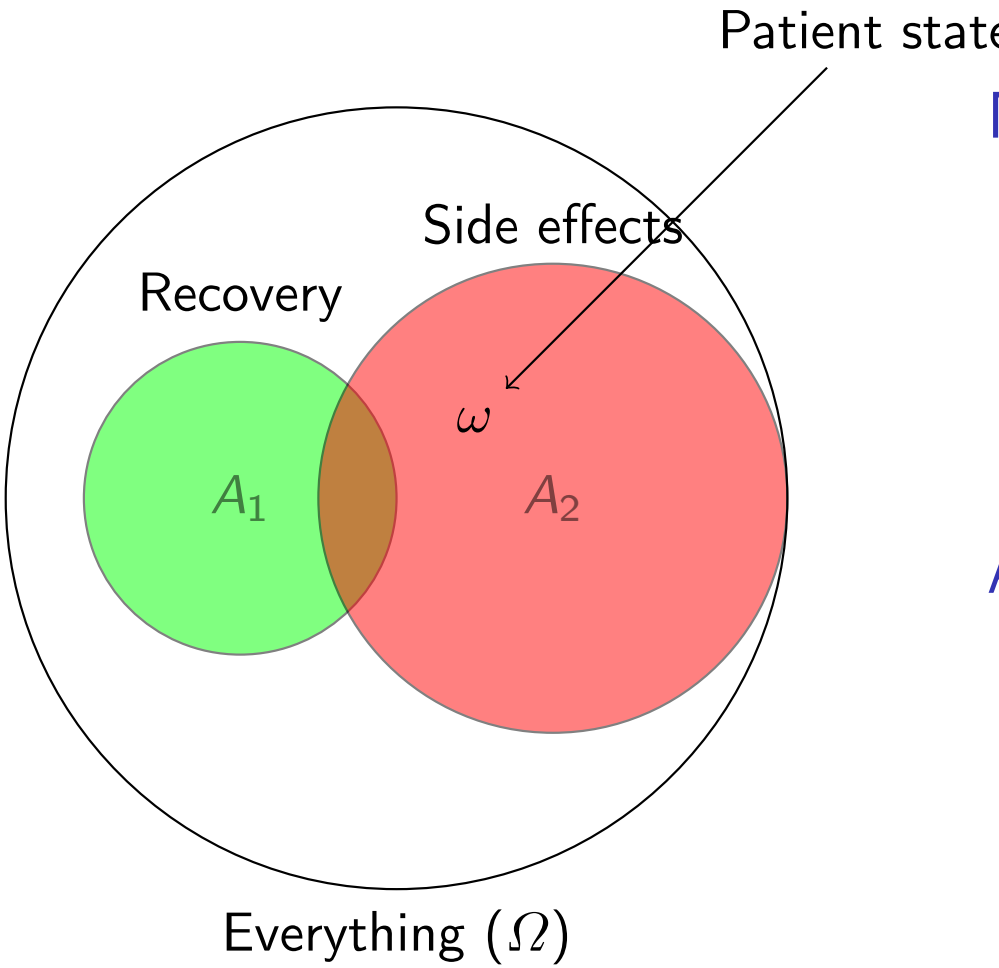
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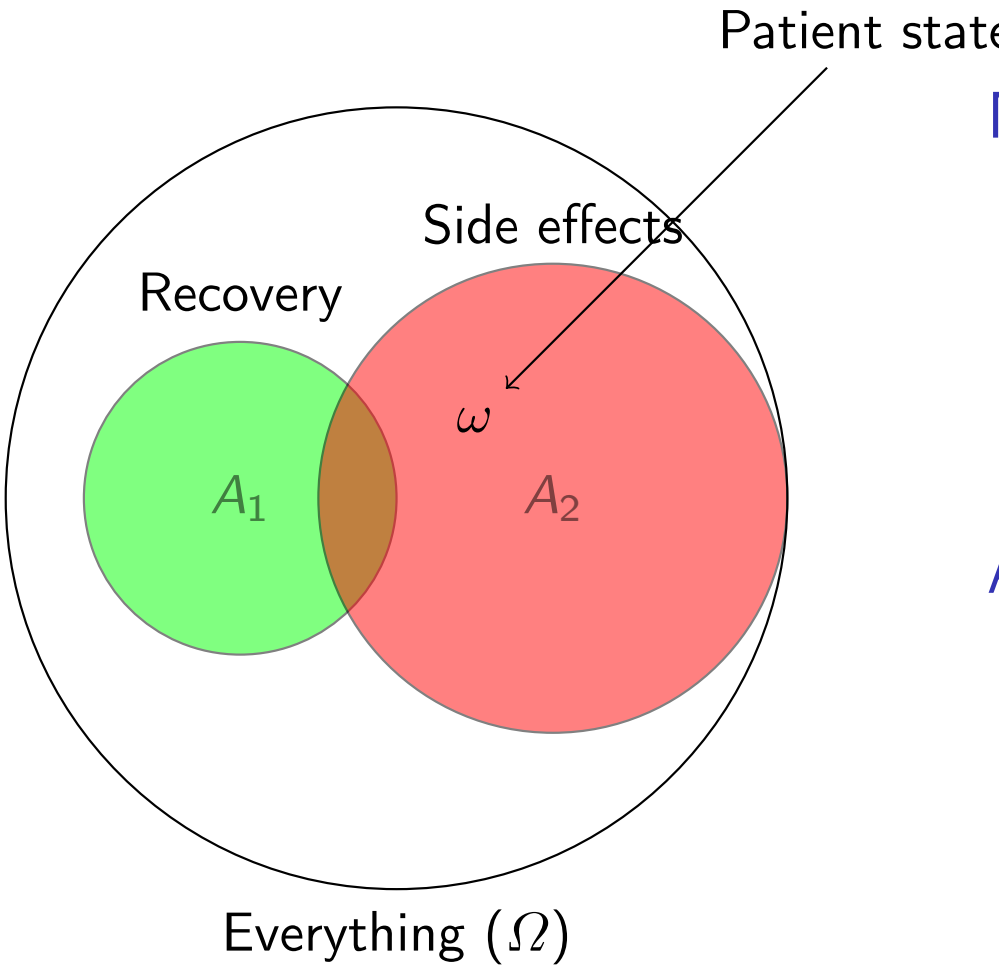
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Axioms

- $P(A) \geq 0$ for any $A \subset \Omega$.

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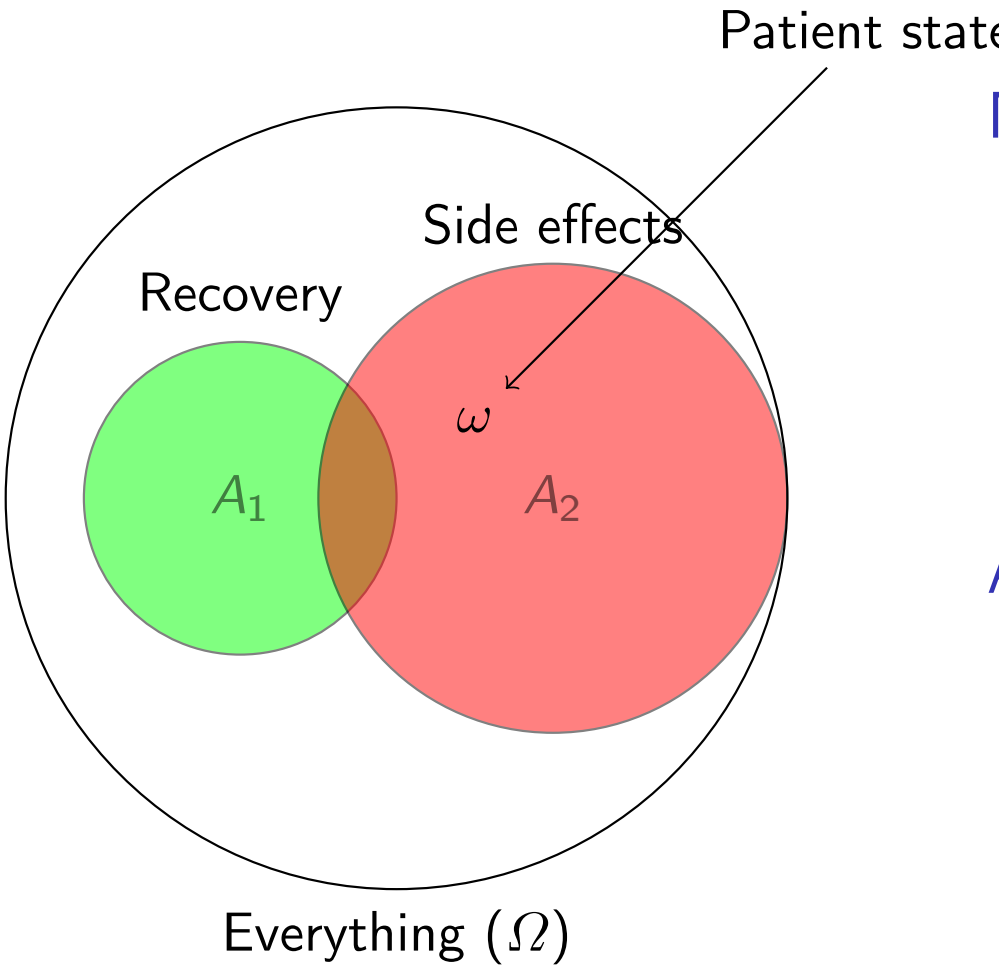
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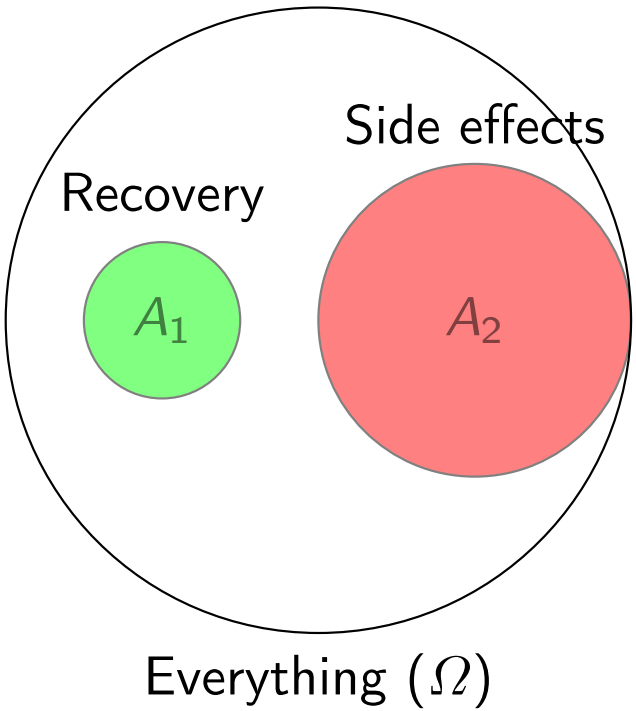
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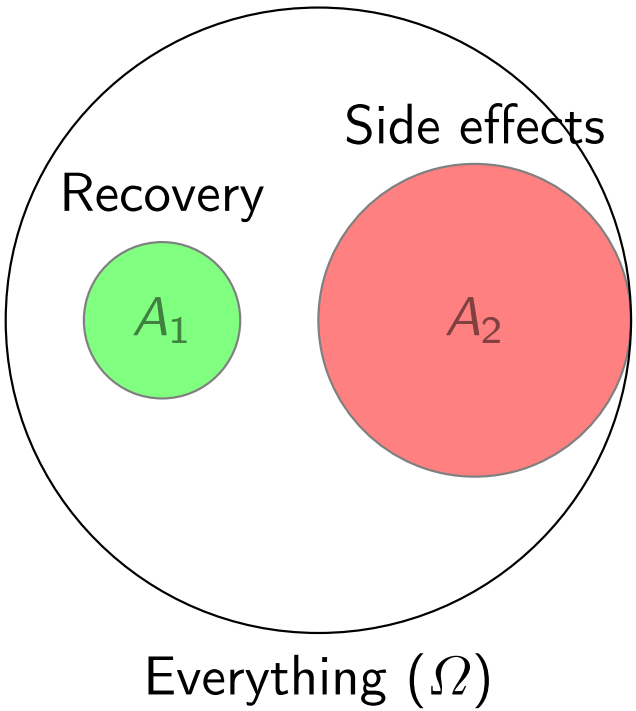
- $P(A) \geq 0$ for any $A \subset \Omega$.
- $P(\Omega) = 1$.
- If $A_1 \cap A_2 = \emptyset$ then $P(A_1 \cup A_2) = P(A_1) + P(A_2)$.

Mutually exclusive events



- $A_1 \cap A_2 = \emptyset$
- Consequently, $P(A_1 \cup A_2) = P(A_1) + P(A_2)$
- So the probability of A_1 or A_2 equals the probability of A_1 plus the probability of A_2.

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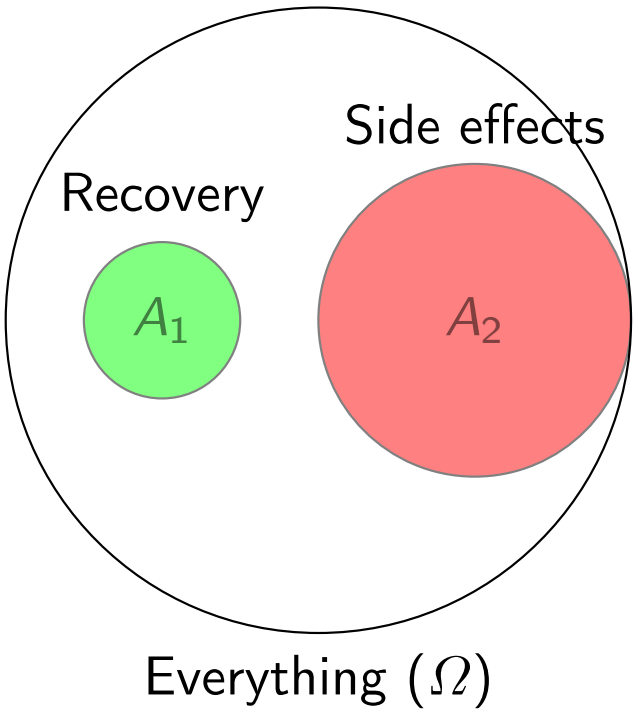


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Which of the following is true?

1. If A_2 happens then A_1 happens.
2. If A_1 happens then A_2 happens.
3. A_1 and A_2 can happen at the same time.
4. A_1 and A_2 cannot happen at the same time.

Mutually exclusive events



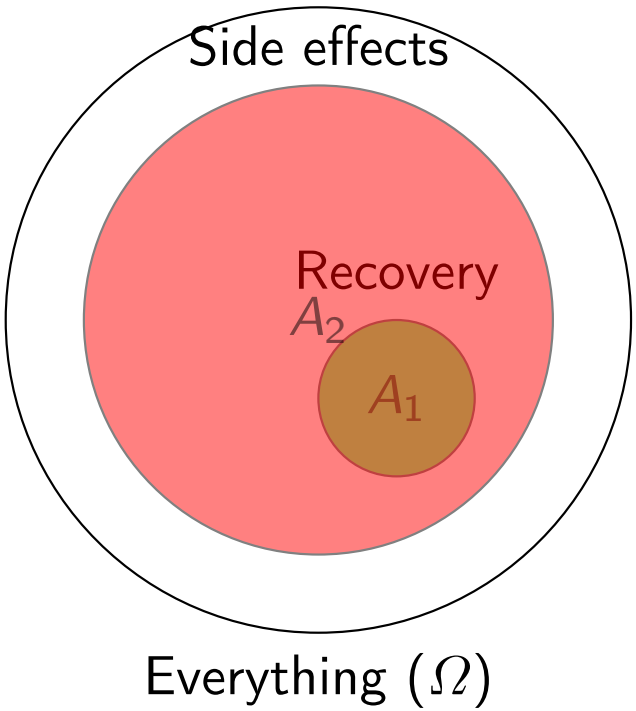
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- (4) is correct.

Implication

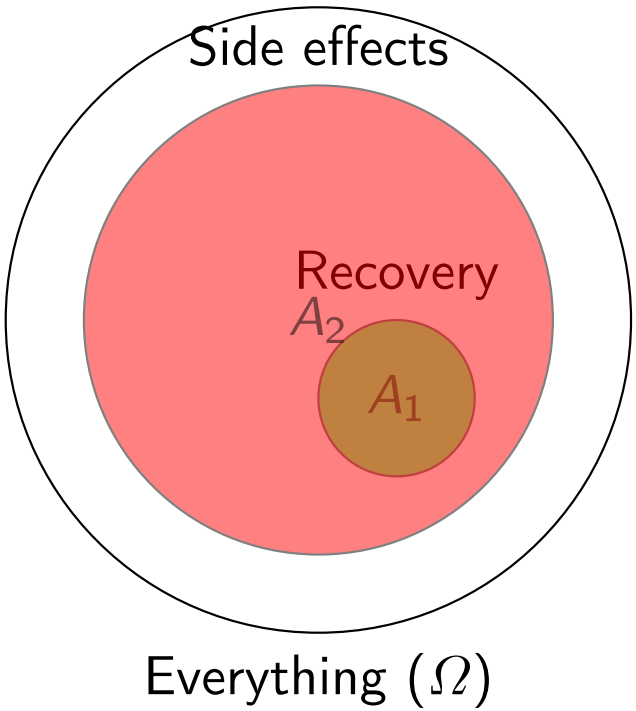


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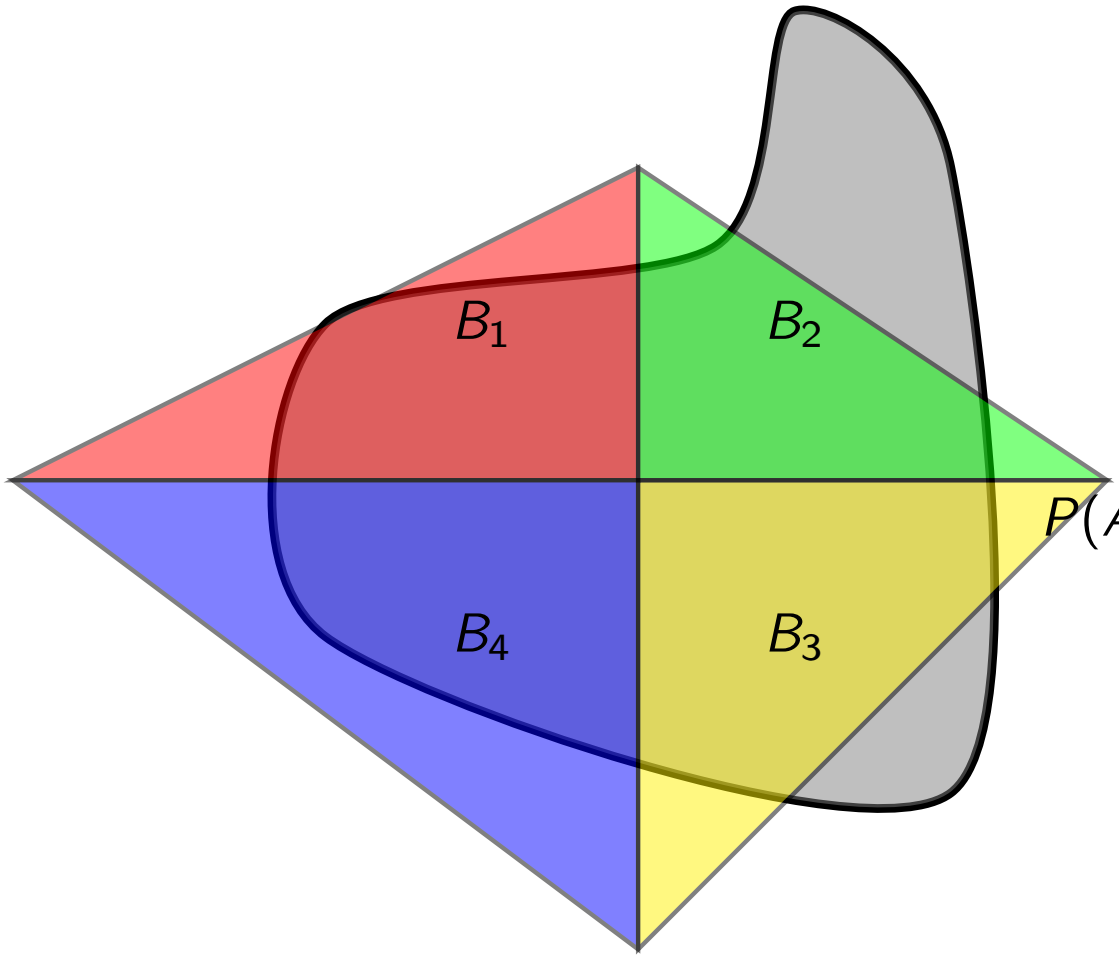
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- (2, 3) are correct

Marginalisation



$$P(B) = P\left(\bigcup_i B_i\right) = \sum_i P(B_i)$$

$$P(A \cap B) = P\left(\bigcup_i (A \cap B_i)\right)$$
$$= \sum_i P(A \cap B_i),$$

Exercises

Implication

- Prove that if $A \subset B$ then $P(A) \leq P(B)$.

Independent vs mutually exclusive events

- Prove that, if A, B are mutually exclusive events with $P(A) > 0, P(B) > 0$ then they are not independent. What is the intuition behind this result?

[Reminder: if A, B are independent then $P(A \cap B) = P(A)P(B)$]

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Bayes theorem

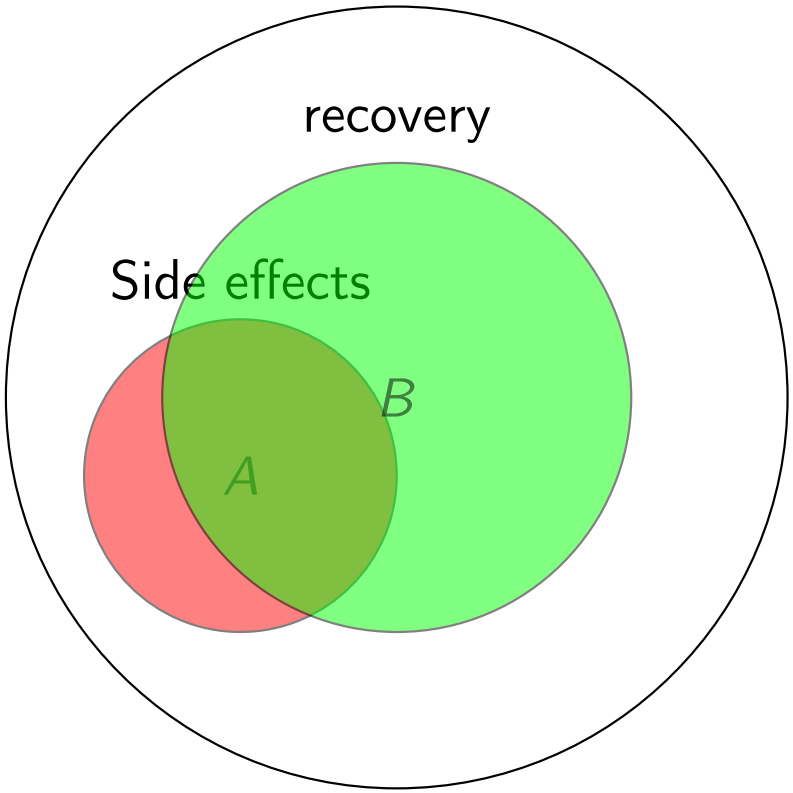
Conditional probability

The probability of A happening if B happens.

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

Since the following is also true:

$$P(B|A) \triangleq \frac{P(A \cap B)}{P(A)}$$



Bayes's theorem

So we can reverse the order of conditioning:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- If you do recover, what is more likely? Side-effects or not?
- If you have side-effects, are you more likely to recover not?

The cards problem

1. Print out a number of cards, with either [A|A], [A|B] or [B|B] on their sides.
2. If you have an A, what is the probability of an A on the other side?
3. Have the students perform the experiment with:
   - 3.1 Draw a random card.
   - 3.2 Count the number of people with A.
   - 3.3 What is the probability that somebody with an A on one side will have an A on the other?
   - 3.4 Half of the people should have an A?

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The prior and posterior probabilities

A	A	2/6	A observed	2/3
A	B	1/6	A observed	1/3
B	A	1/6		
B	B	2/6		

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- This is a purely subjective measure!

DNA test

- Let us now do a DNA test on the suspect

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DNA test properties

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Everybody is a suspect

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- What is your belief that the people with the positive test are guilty?

Explanation

- Prior: $P(H_i)$. How much do we believe in H_i.
- Likelihood $P(D|H_i)$. How likely is D if H_i is true.

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- Posterior $P(H_i|D)$: How likely is H_i given the data

$$P(H_i|D) = \frac{P(D \cap H_i)}{P(D)} = \frac{P(D|H_i)P(H_i)}{P(D)}$$

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$$P(H_i|D) = \frac{P(D \cap H_i)}{P(D)} = \frac{P(D|H_i)P(H_i)}{P(D)}$$

- Obtaining the marginal probability:

$$P(D) = P(D \cap H_0) + P(D \cap H_1) = P(D|H_0)P(H_0) + P(D|H_1)P(H_1)$$

- Posterior:

$$P(H_0|D) = \frac{P(D|H_0)P(H_0)}{P(D|H_0)P(H_0) + P(D|H_1)P(H_1)}$$

- Assuming $P(D|H_1) = 1$, and setting $P(H_0) = q$, this gives

$$P(H_0|D) = \frac{0.1q}{0.1q + 1 - q} = \frac{q}{10 - 9q}$$

- The posterior can always be updated with more data!

The Covid Test

- You are feeling sick. Two possibilities:
  - H_0: You do not have COVID.
  - H_1: You have COVID.
- You perform a rapid test, which has detects COVID 80% of the time.

$$P(T|H_1) = 0.8.$$

- It falsely says you have COVID 10 % of the time.

$$P(T|H_0) = 0.1.$$



- Throw two dice dice and fill in the result in the google sheet to get your test result.
- Select your own prior probability $P(H_1)$ of having covid. What is the posterior probability $P(H_1|T)$

Python example

```
def get_posterior(prior, data, likelihood):
        marginal = prior*likelihood[data][0]
         + (1 - prior)*likelihood[data][1]
        posterior = prior*likelihood[data][0] / marginal
        return posterior

import numpy as np
prior = 0.9 # Pr(H0)
likelihood = np.zeros([2, 2])
likelihood[0][0] = 0.9 # Pr(F|H0)
likelihood[1][0] = 0.1 # Pr(T|H0)
likelihood[0][1] = 0 # Pr(F|H1)
likelihood[1][1] = 1 # Pr(T|H1)
data = 1
return get_posterior(prior, data, likelihood)
```

Types of hypothesis testing problems

Simple Hypothesis Test

Example: DNA evidence, Covid tests

- Two hypothesese H_0, H_1
- $P(D|H_i)$ is defined for all i

Multiple Hypotheses Test

Example: Model selection

- H_i: One of many mutually exclusive models
- $P(D|H_i)$ is defined for all i

Null Hypothesis Test

Example: Are men's and women's heights the same?

- H_0: The 'null' hypothesis
- $P(D|H_0)$ is defined
- The alternative is undefined

Pitfalls

Problem definition

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The garden of many paths

- Having a huge hypothesis space
- Selecting the relevant hypothesis after seeing the data

Class feedback



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- Model family $\{P_\theta | \theta \in \mathcal{H}\}$

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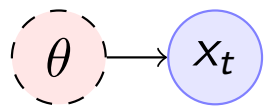
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Example (Bernoulli model)

- $x_t \in \{0, 1\}$, $\theta \in [0, 1]$

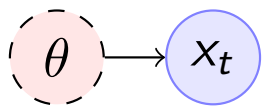


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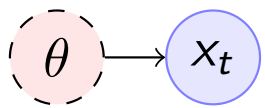


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- $P_\theta(1) = \theta$

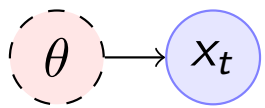


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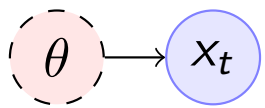


Probabilistic models

- Model family $\{P_\theta | \theta \in \Theta\}$
- Data $x_t \mid \theta^* \sim P_{\theta^*}$ for some $\theta^* \in \Theta$ and $t = 1, \ldots, T$.
- How can we estimate the correct θ?
- How can we predict a new data point?

Example (Bernoulli model)

- $x_t \in \{0, 1\}$, $\theta \in [0, 1]$
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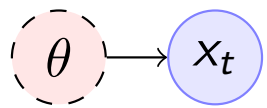


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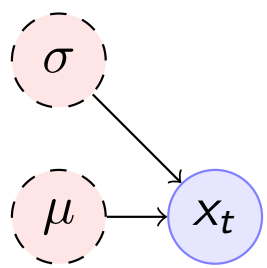
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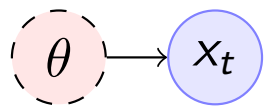


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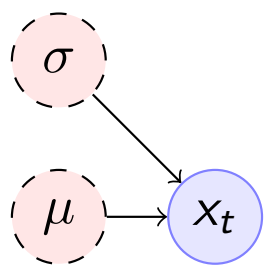
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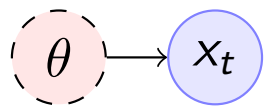


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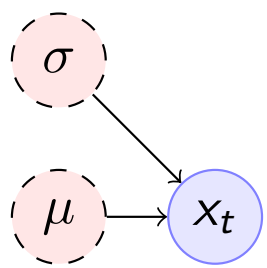
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Notes on Probabilistic Models. The model family describes which sets of models we are interested in. For example, it can be the set of Gaussian models, the set of Bernoulli models, or anything else. θ is usually called the parameter.

Each model θ assigns probabilities $P_\theta(x)$ to possible x. You an alternative write this as $\mathbb{P}(x \mid \theta)$, to mean the conditional probability of x given θ.

We use x_t to emphasize that we have multiple samples, which in this case are i.i.d. In fact,

$$P_\theta(x_1, \ldots, x_T) = \prod_{t=1}^{T} P_\theta(x_1, \ldots, x_T)$$

or in the standard notation

$$\mathbb{P}(x_1, \ldots, x_T \mid \theta) = \prod_{t=1}^{T} \mathbb{P}(x_1, \ldots, x_T \mid \theta)$$

Maximum likelihood (ML) inference

- Family $\{P_\theta | \theta \in \Theta\}$
- Data x with likelihood $P_\theta(x)$ for each parameter value θ.
- $\theta_{\mathrm{ML}}(x) = \arg\max_\theta P_\theta(x)$

Example (Bernoulli model)

- $x_t \in \{0, 1\}$, for $t \in [T]$, $\theta \in [0, 1]$

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- Define $s_T = \sum_{t=1}^T x_t$.

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- Show that $\theta_{\mathrm{ML}}(x) = s_T / T$.

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- Define $s_T = \sum_{t=1}^T x_t$.
- Show that $\theta_{\mathrm{ML}}(x) = s_T / T$.
- What is the problem with this estimate?

Notes on Maximum Likelihood. In maximum likelihood inference, we have a model family as before, and some data. The idea is to find the parameter that best fits the data. The natural way to do so is to use the probabilities: find θ maximising the probability of the data. (This is also called the likelihood of the parameter), i.e.

$$\theta_{\mathrm{ML}} = \arg\max_\theta P_\theta(x).$$

Reminder that the argument maximising a function results in the maximum value, i.e.

$$P_{\theta_{\mathrm{ML}}}(x) = \max_\theta P_\theta(x) \geq P_{\theta'}(x) \forall \theta'.$$

For the Bernoulli model, consider what happens when we throw the coin only once and it comes heads. What is our estimate then? Is it reasonable? Can we use the estimate to make predictions?

Gaussian model

- $x_t \in \mathbb{R}$, for $t \in [T]$, $\mu \in \mathbb{R}, \sigma \in \mathbb{R}_+$
- $x_t \mid \mu, \sigma \sim \mathrm{Normal}(\mu, \sigma)$
- We can show that the maximum likelihood parameter estimate is:

$$\hat{\mu}_{\mathrm{ML}}(x) = \frac{1}{T}\sum_{t=1}^{T} x_t, \qquad \hat{\sigma}^2_{\mathrm{ML}}(x) = \frac{1}{T}\sum_{t=1}^{T} \left[x_t - \hat{\mu}_{\mathrm{ML}}(x)\right]^2.$$

- To prove this, (1) write the log-likelihood (2) differentiate with respect to each parameter separately (3) set the derivative to zero.
- Does this estimator for σ^2 have a problem analogous to the Bernoulli one above? (Hint: what happens when $T = 1$?)

Maximum a posteriori (MAP) inference

- Family $\{P_\theta | \theta \in \Theta\}$
- Data x with likelihood $P_\theta(x)$ for each parameter value θ.
- Prior $\phi(\theta)$
- $\theta_{\mathrm{MAP}}(x) = \arg\max_\theta P_\theta(x)\phi(\theta)$
- Experiment with the prior for the Bernoulli model.

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Notes on Maximum A Posteriori. In maximum a posteriori inference, we have a model family as before, and some data. However, we may think of some models as inherently more or less likely than other models. For that reason, we temper or regulate our estimate with a prior distribution (or density) ϕ.

For the Bernoulli model, think again about what happens when we throw the coin only once and it comes heads. Obviously the conclusion that the coin will always come up as heads afterwards is ridiculous. For that reason, we can consider a prior ϕ which places 0 weight on $\theta = 0$ or $\theta = 1$. In fact, something which places more weight towards fair coins seems more reasonable. A good choice is:

$$\phi(\theta) \propto \theta(1-\theta).$$

Here we use proportionality, this must be scaled so that $\int_0^1 \phi(\theta) d\theta = 1$. However, this does not matter, as in MAP inference we calculate

$$\arg\max_\theta \phi(param) P_\theta(x)$$

In this case, multiplication by a positive constant does not change the maximising argument.

Bayesian Inference

- Model family $\{P_\theta | \theta \in \Theta\}$
- Each model θ assigns probabilities $P_\theta(x)$ to possible $x \in X$.
- We also have a (subjective) prior distribution ϕ over the parameters.
- Given x, we calculate the posterior distribution:

$$\phi(\theta|x) = \frac{P_\theta(x)\phi(\theta)}{\sum_{\theta' \in \Theta} P_{\theta'}(x)\phi(\theta')}, \qquad \text{(finite } \Theta\text{, } \phi \text{ is a probability)}$$

$$\phi(\theta|x) = \frac{P_\theta(x)\phi(\theta)}{\int_\Theta P_{\theta'}(x)\phi(\theta')d\theta'}, \qquad \text{(continuous } \Theta\text{, } \phi \text{ is a density)}$$

$$\phi(B|x) = \frac{\int_B P_{\theta'}(x)d\phi(\theta)}{\int_\Theta P_{\theta'}(x)d\phi(\theta)}, \qquad B \subset \Theta \qquad \text{(arbitrary } \Theta\text{, } \phi \text{ is a measure)}$$

Alternative notation for different probability spaces

- The prior $\phi(\theta) = \mathbb{P}(\theta)$ and posterior $\phi(\theta \mid x) = \mathbb{P}(\theta \mid x)$ belief.
- The likelihood $P_\theta(x) = \mathbb{P}(x \mid \theta)$
- The marginal $\mathbb{P}_\phi(x) = \sum_\theta P_\theta(x)\phi(\theta)$.

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Notes on Bayesian Inference. The model family describes which sets of models we are interested in. For example, it can be the set of Gaussian models, the set of Bernoulli models, or anything else. θ is usually called the parameter.
Each model θ assigns probabilities $P_\theta(x)$ to possible x.
Since we do not know θ, we assign a subjective prior distribution ϕ to every possible value of $\theta \in \Theta$. The mathematics gets a bit more convoluted here, because there are many different cases for what Θ can be.

- If Θ only has a finite number of values, then ϕ is a probability mass function, so that $\sum_{\theta \in \Theta} \phi(\theta) = 1$ and $\phi(\theta) \in [0, 1]$.
- If Θ is continuous (e.g. the interval $[0, 1]$) then it ϕ is a probability density function, so that $\int_\Theta \phi(\theta) d\theta = 1$, and $\phi(\theta) \geq 0$. However, $\phi(\theta)$ can be arbitrarily large!
- In the most general case, we can define ϕ to be a probability measure. Then it is a function on sets (like in basic probability). Now, however, the integral is written: $\int_\Theta d\phi(\theta) = 1$, and we have that $\phi(B) \in [0, 1]$ for every subset $B \subset \Theta$.

The reason for the alternative notation is to conceptually separate the model's probabilities for the data $P_\theta(x) \triangleq \mathbb{P}(x \mid \theta)$ from our prior and posterior beliefs $\phi(\theta) \triangleq \mathbb{P}(\theta)$ and $\phi(\theta \mid x) \triangleq \mathbb{P}(\theta \mid x)$ about the model. However, you can use whichever notation feels best for you, as long as you are consistent.

Probabilistic machine learning

- Model family $\{P_\theta | \theta \in \mathcal{H}\}$

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- Model family $\{P_\theta | \theta \in \mathcal{H}\}$
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- Model family $\{P_\theta | \theta \in \Theta\}$
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Maximum likelihood approach

- Model selection: $\theta^*_{ML}(x) = \arg\max_\theta P_\theta(x)$.
- Model prediction: $P_{\theta^*_{ML}(x)}(x_{t+1})$

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Bayesian approach

- Posterior calculation: $\phi(\theta|x) = P_\theta(x)\phi(\theta)/\mathbb{P}_\phi(x)$
- Model prediction: $\mathbb{P}_\phi(x_{t+1}|x) = \sum_\theta P_\theta(x_{t+1})\phi(\theta|x)$

Differences between approaches

Maximum likelihood approach

- Ignores model complexity
- Is an optimisation problem

Maximum a posteriori approach

- Regularises model selection using the prior
- Can be seen as solving the optimisation problem

$$\max_{\theta} \ln P_{\theta}(x) + \ln \phi(\theta),$$

where the prior term $\ln \phi(\theta)$ acts as a regulariser.

Bayesian approach

- Does not select a single model
- Averages over all models according to their fit and the prior
- Does not result in an optimisation problem.

The n-meteorologists problem

- Consider n meteorological stations $\{\mu\}$ predicting rainfall.
- $x_t \in \{0, 1\}$ with $x_t = 1$ if it rains on day t.
- We have a prior distribution $\phi(\mu)$ for each station.
- At time t, station μ makes as a prediction $P_\mu(x_{t+1}|x_1, \ldots, x_t)$
- We observe x_{t+1} and calculate the posterior $\phi(\mu|x_1, \ldots, x_t, x_{t+1})$.

The marginal distribution

To take into account all stations, we can marginalise:

$$\mathbb{P}_\phi(x_{t+1} \mid x_1, \ldots x_t) = \sum_\mu P_\mu(x_{t+1}|x_t)\phi(\mu)$$

The posterior

- Show that

$$\phi(\mu \mid x_1, \ldots, x_{t+1}) = \frac{P_\mu(x_t \mid x_1, \ldots, x_t)\phi(\mu|x_1, \ldots, x_t)}{\sum_{\mu'} P_{\mu'}(x_t \mid x_1, \ldots, x_t)\phi(\mu'|x_1, \ldots, x_t)}$$

- How would you implement an ML or a MAP solution to this problem?

Sufficient statistics

A statistic f

This is any function $f : X \to S$ where

- X is the data space
- S is an arbitrary space

Example statistics for $X = \mathbb{R}^*$ (the set of all real-valued sequences)

- The sample mean of a sequence $1/T \sum_{t=1}^{T} x_t$
- The total number of samples T

Sufficient statistic

f is sufficient for a family $\{P_\theta : \theta \in \Theta\}$ when

$$f(x) = f(x') \Rightarrow P_\theta(x) = P_\theta(x') \forall \theta \in \Theta.$$

If there exists a finite-dimensional sufficient statistic, Bayesian and ML learning can be done in closed form within the family.

Conjugate priors

Consider a parametrised family of priors Φ on Θ and a distribution family $\{P_\theta\}$
The pair is conjugate if, for any prior $\phi \in \Phi$, and any observation x, there exists $\phi' \in \Phi$ such that $\phi'(\theta) = \phi(\theta|x)$

Standard Parametric conjugate families

Prior	Likelihood	Parameters θ	Observations x
Beta	Bernoulli	$[0, 1]$	$\{0, 1\}^T$
Multinomial	Dirichlet	$\mathbb{\Delta}^n$	$\{1, \ldots, n\}^T$
Gamma	Normal	$\mathbb{R}, \mathbb{R}$	$\mathbb{R}^T$
Wishart	Normal	$\mathbb{R}^n, \mathbb{R}^{n\times n}$	$\mathbb{R}^{n\times T}$

The Simplex $\mathbb{\Delta}^n = \{\boldsymbol{\theta} \in [0, 1]^n : \|\boldsymbol{\theta}\|_1\}$ is the set of all n-dimensional probability vectors.

Extensions

- Discrete Bayesian Networks.
- Linear-Gaussian Models (i.e. Bayesian linear regression)
- Gaussian Processes.

Beta-Bernoulli



Definition of the Bernoulli distribution

If $x_t \mid \theta \sim \text{Bernoulli}(\theta)$. $\theta \in [0, 1]$, $x_t \in \{0, 1\}$ and:

$$P_\theta(x_t = 1) = \theta$$

Definition of the Beta density

If $\theta \sim \text{Beta}(\alpha_1, \alpha_0)$, $\alpha_0, \alpha_1 > 0$ and

$$p(\theta|\alpha_1, \alpha_0) \propto \theta^{\alpha_1 - 1}(1 - \theta)^{\alpha_0 - 1}$$

Beta-Bernoulli conjugate pair

- $\theta \sim \text{Beta}(\alpha_1, \alpha_0)$.
- $x_t \mid \theta \sim \text{Bernoulli}(\theta)$.

Then, for any $x = x_1, \ldots, x_T$, the posterior distribution is

- $\theta \mid x \sim \text{Beta}(\alpha_1 + \sum_t x_t, \alpha_0 + T - \sum_t x_t)$.

Dirichlet-Multinomial



Definition of the Multinomial distribution

If $x_t \mid \boldsymbol{\theta} \sim \mathrm{Mult}(\boldsymbol{\theta})$, with $\theta \in \mathbb{\Delta}^n$ and $x_t \in \{1, \ldots, n\}$ and:

$$P_{\boldsymbol{\theta}}(x_t = i) = \theta_i$$

Definition of the Dirichlet density

If $\boldsymbol{\theta} \sim \mathrm{Dir}(\boldsymbol{\alpha})$, with $\boldsymbol{\alpha} \in \mathbb{R}_+^n$ then $p(\theta|\boldsymbol{\alpha}) \propto \prod_i \theta_i^{\alpha_i - 1}$

Dirichlet-Multinomial conjugate pair

- $\theta \sim \mathrm{Dir}(\boldsymbol{\alpha})$.
- $x_t \mid \theta \sim \mathrm{Bernoulli}(\boldsymbol{\theta})$.

Then, for any $x = x_1, \ldots, x_T$, the posterior distribution is

- $\theta \mid x \sim \mathrm{Dir}(\boldsymbol{\alpha} + \boldsymbol{s_T})$, where $s_{T,i} = \sum_{t=1}^{T} \mathbb{I}\{x_t = i\}$,

Introduction

Conditional Probability and the Theorem of Bayes

Simple Bayesian hypothesis testing

Bayesian Inference

Exercises

Probability

Normal distribution

- Find the MLE for the normal distribution. What is the sufficient statistic?